

Performance Estimation Problems (PEPs)

Systematic analysis and design of first-order optimization algorithms
via the performance estimation framework

François Glineur, Adrien Taylor



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| Designing algorithms

A “generic” first-order method

$$w_1 = w_0 - \alpha_{1,0} \nabla f(w_0)$$

$$w_2 = w_1 - \alpha_{2,0} \nabla f(w_0) - \alpha_{2,1} \nabla f(w_1)$$

$$w_3 = w_2 - \alpha_{3,0} \nabla f(w_0) - \alpha_{3,1} \nabla f(w_1) - \alpha_{3,2} \nabla f(w_2)$$

$\vdots$

$$w_n = w_{n-1} - \sum_{i=0}^{n-1} \alpha_{n,i} \nabla f(w_i),$$

(FOM)

for some coefficients $\{\alpha_{i,j}\}$. Generic **non-adaptive** first-order method.

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for some coefficients $\{\alpha_{i,j}\}$. Generic **non-adaptive** first-order method.

How to choose $\{\alpha_{i,j}\}$?

- ◇ pick a performance criterion, for instance $\frac{\|w_n - w_\star\|^2}{\|w_0 - w_\star\|^2}$,
- ◇ solve the minimax (minimize worst-case): $\min_{\{\alpha_{i,j}\}_{i,j}} \max_{f \in \mathcal{F}, \{w_i\}} \frac{\|w_n - w_\star\|^2}{\|w_0 - w_\star\|^2}.$

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- ◇ unconstrained quadratic programming:

first-order methods $\Leftrightarrow$ polynomials $\rightarrow$ optimal algorithms via optimal polynomials.^{1,2,3}

¹Golub, Varga (1961). "Chebyshev semi-iterative methods, successive overrelaxation iterative methods, and second order Richardson iterative methods."

²Polyak (1987). "Introduction to optimization."

³Nemirovski (1995). "Information-based complexity of convex programming." (Lecture notes)

| Traditional design

Algorithm design (with guarantees):

- ◇ unconstrained quadratic programming:
first-order methods $\Leftrightarrow$ polynomials $\rightarrow$ optimal algorithms via optimal polynomials.^{1,2,3}
- ◇ Beyond quadratics: traditionally more “hand-crafted”
 - Analogy with conjugate gradients.^{4,5}
 - Lyapunov function-type analyses.^{6,7}

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⁴Nemirovski (1982). “Orth-method for smooth convex optimization.” (in Russian)

⁵Narkiss, Zibulevsky (2005). “Sequential subspace optimization method for large-scale unconstrained problems.”

⁶Nesterov (1983). “A method for solving the convex programming problem with convergence rate $O(1/k^2)$.”

⁷Beck, Teboulle (2009). “A fast iterative shrinkage-thresholding algorithm for linear inverse problems.”

| Design problem

How to solve the design problem (or proxy of it)?

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- ◇ Convex relaxations,
- ◇ analogies (e.g., with conjugate gradient methods)

$$w_{k+1} \in \operatorname{argmin}_{x \in \mathbb{R}^d} \{f(w) : w \in w_0 + \operatorname{span}\{\nabla f(w_0), \dots, \nabla f(w_k)\}\},$$

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- ◇ brutal approaches.

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Performance estimation problem with step-size optimization (with $n = 1$):

$$\begin{aligned} \min_{\alpha_{1,0}} \max_{G, F} \quad & G_{1,1} + \alpha_{1,0}^2 G_{2,2} - 2\alpha_{1,0} G_{1,2} \\ \text{subject to} \quad & F + \frac{L\mu}{2(L-\mu)} G_{1,1} + \frac{1}{2(L-\mu)} G_{2,2} - \frac{L}{L-\mu} G_{1,2} \leq 0 \\ & -F + \frac{L\mu}{2(L-\mu)} G_{1,1} + \frac{1}{2(L-\mu)} G_{2,2} - \frac{\mu}{L-\mu} G_{1,2} \leq 0 \\ & G_{1,1} = 1 \\ & G \succcurlyeq 0. \end{aligned}$$

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“Simple” minimization problem by dualizing inner maximization.

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$$\text{Symmetric } M = \begin{pmatrix} A & B \\ B^T & C \end{pmatrix}$$

$$\text{if } C \succ 0: M \succeq 0 \Leftrightarrow A - BC^{-1}B^T \succeq 0$$

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For $n = 1$, **optimizing over step-size** $\alpha_{1,0}$ remains convex!

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When $n = 2$, the problem becomes

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for some $S_{1,1}, S_{1,2}, \dots, S_{3,3}$ (functions of $\tau, \lambda_1, \dots, \lambda_6$ and $\{\alpha_{i,j}\}$).

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For instance:

$$\begin{aligned} S_{1,2} &= -\frac{L\lambda_3 - 2(L-\mu)\alpha_{2,0} + \mu\lambda_1 + L\mu(\lambda_2 + \lambda_5)\alpha_{1,0}}{L-\mu} \\ S_{2,2} &= \frac{-2(\mu\lambda_6 + L\lambda_4)\alpha_{1,0} - 2(L-\mu)\alpha_{2,0}^2 + L\mu(\lambda_2 + \lambda_4 + \lambda_5 + \lambda_6)\alpha_{1,0}^2 + \lambda_1 + \lambda_3 + \lambda_4 + \lambda_6}{L-\mu} \end{aligned}$$

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convex in some step-sizes ($\alpha_{2,0}$ and $\alpha_{2,1}$) but not in the others.

| Hands-on

Minimize an L -smooth μ -strongly convex function.

- ◇ Goal 1: design an optimized one-step gradient method

$$x_1 = x_0 - \alpha_0 \nabla f(x_0),$$

for optimizing the worst-case performance $\frac{\|x_1 - x_\star\|^2}{\|x_0 - x_\star\|^2}$.

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- ◇ Goal 2: design an optimized two-step gradient method

$$x_1 = x_0 - \alpha_0 \nabla f(x_0),$$

$$x_2 = x_1 - \alpha_1 \nabla f(x_1),$$

for optimizing the worst-case performance $\frac{\|x_2 - x_\star\|^2}{\|x_0 - x_\star\|^2}$.

| Hands-on

Minimize an L -smooth μ -strongly convex function.

- ◇ Goal 1: design an optimized one-step gradient method

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- ◇ Goal 2: design an optimized two-step gradient method

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$$x_2 = x_1 - \alpha_1 \nabla f(x_1),$$

for optimizing the worst-case performance $\frac{\|x_2 - x_\star\|^2}{\|x_0 - x_\star\|^2}$.

Tutorial material:

<https://github.com/PerformanceEstimation/Tutorial-SIAM-OP26>

| A few instructive examples

Design first-order methods via PEPs:

- ◇ Drori, Teboulle (2014). "Performance of first-order methods for smooth convex minimization: a novel approach."
- ◇ Kim, Fessler (2016). "Optimized methods for smooth convex optimization."
- ◇ Van Scoy, Freeman, Lynch (2017). "The fastest known globally convergent first-order method for minimizing strongly convex functions."
- ◇ Kim, Fessler (2021). "Optimizing the efficiency of first-order methods for decreasing the gradient of smooth convex functions."
- ◇ Altschuler, Parrilo (2023). "Acceleration by Stepsize Hedging I: Multi-Step Descent and the Silver Stepsize Schedule."
- ◇ Jang, Das Gupta, Ryu (2025). "Computer-assisted design of accelerated composite optimization methods: OptISTA."
- ◇ Bok, Altschuler (2025). "Optimized methods for composite optimization: a reduction perspective."

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- ◇ Bok, Altschuler (2025). "Optimized methods for composite optimization: a reduction perspective."

... including "brutal" examples:

- ◇ Grimmer (2024). "Provably faster gradient descent via long steps."
- ◇ Gupta, Van Parijs, Ryu (2024). "Branch-and-Bound Performance Estimation Programming: A Unified Methodology for Constructing Optimal Methods."
- ◇ Kamri, Hendrickx, Glineur (2025). "Numerical Design of Optimized First-Order Algorithms."